

# Dynamic Personality Adaptation in Large Language Models via State Machines

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## Website



pertrain.ai

## Full Paper



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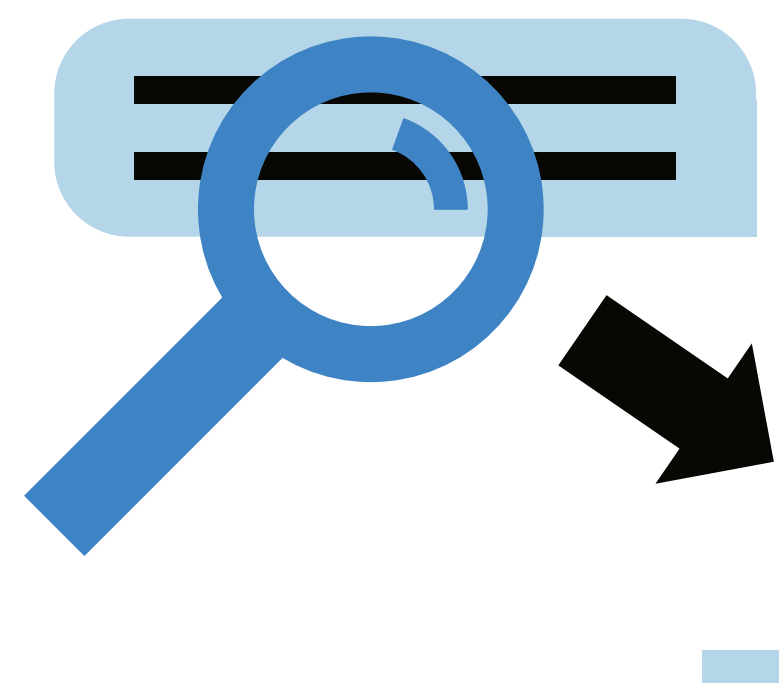
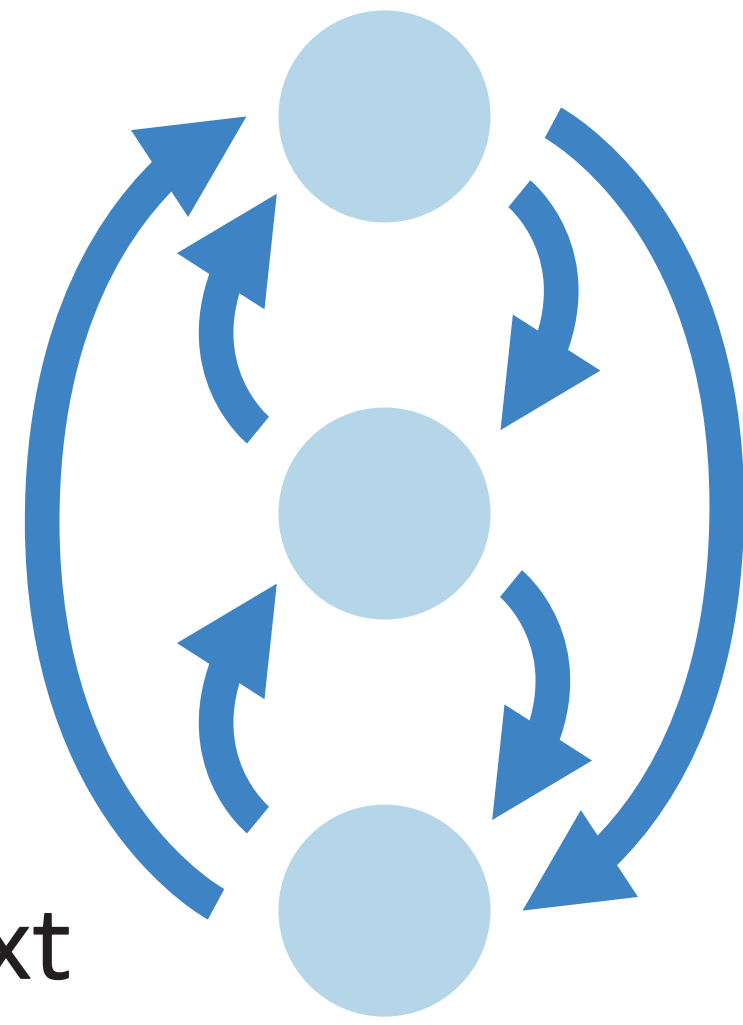
## Code



uni.ms/ycqgz

## Idea

- Model-agnostic framework for dynamic personality simulation
- State machines to represent latent personality states
- Transition probabilities that adapt dynamically to conversation context

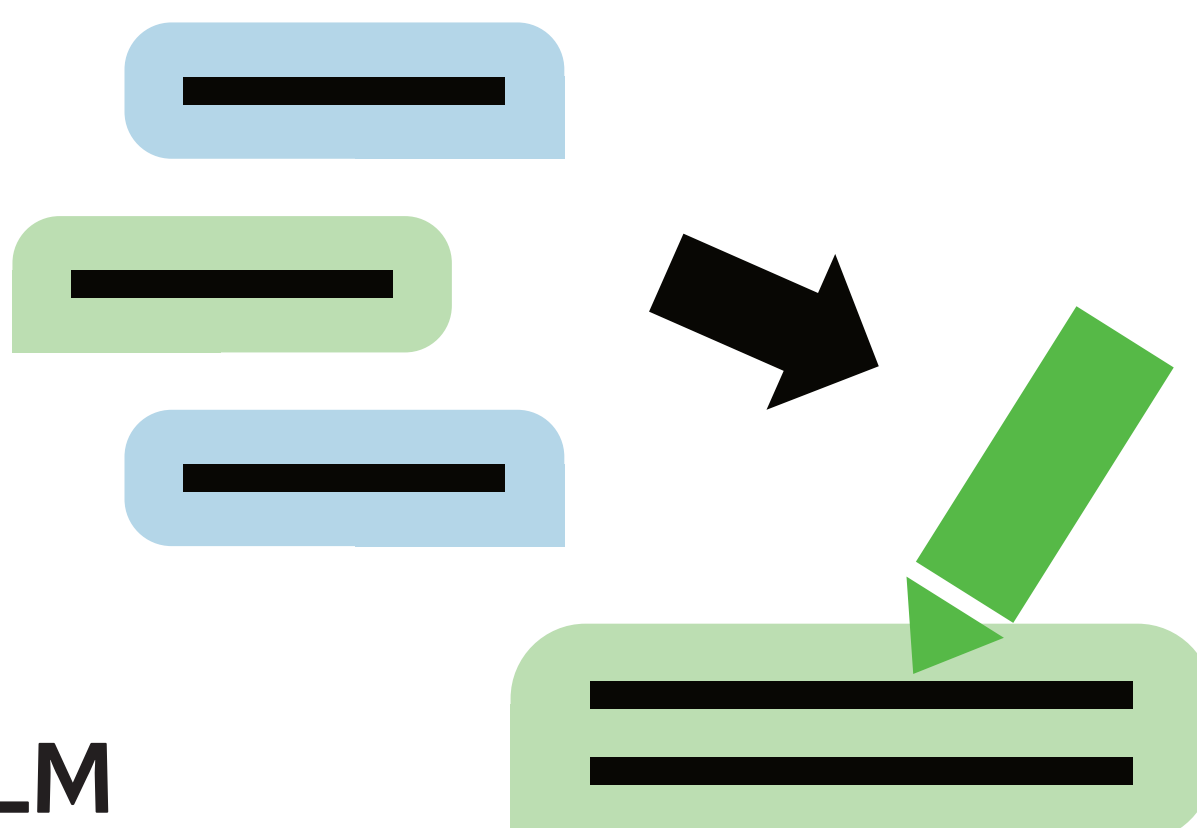


## Analyzer

- Continuous personality scoring
- Rate each message
- Flexible model

## Generator

- Adjust messages according to scenario context and current personality state
- Compatible with any LLM



What's this all about? Looks boring...

We adjust the **personality expression** of LLMs **dynamically** during an interaction. Therefore, it can switch to angry or friendly behaviour depending on the users own expression.

That's great, but now explain how that relates to real human behavior.

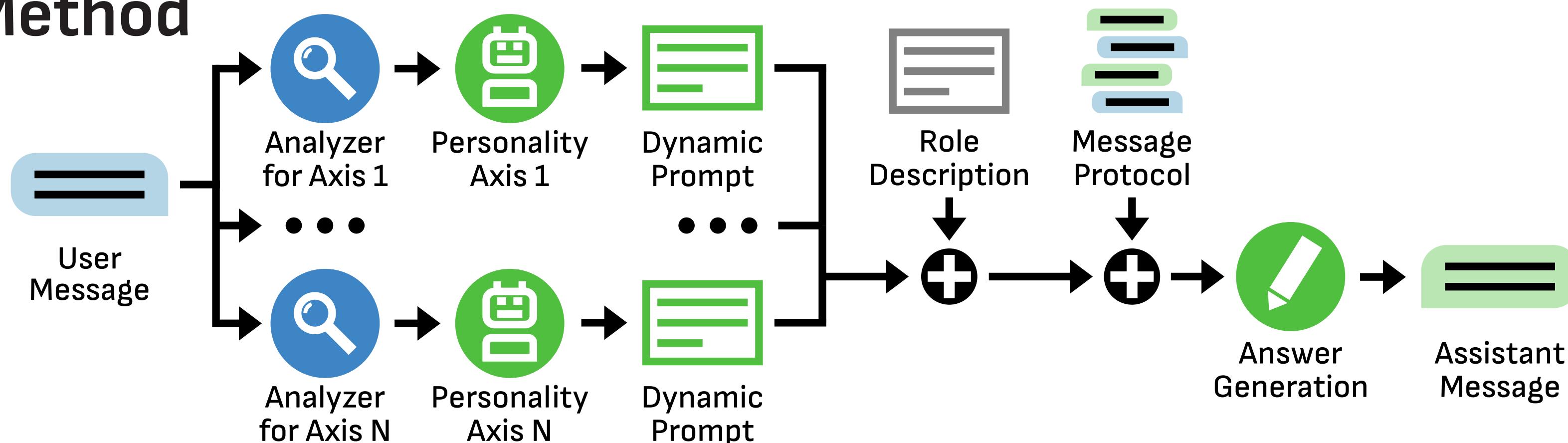
We used the **Interpersonal Circumplex** model as the backbone of our system. It describes behaviour in interpersonal interactions using two **personality axes**. But our system can also support other models flexibly.



Thanks for all the explanations. But are there any real-world use cases?

Around **150 students** have already used our tool for **self-learning** and **in-person learning** in medical education. Students were able to simulate **patient interactions** and practise their **communication skills**.

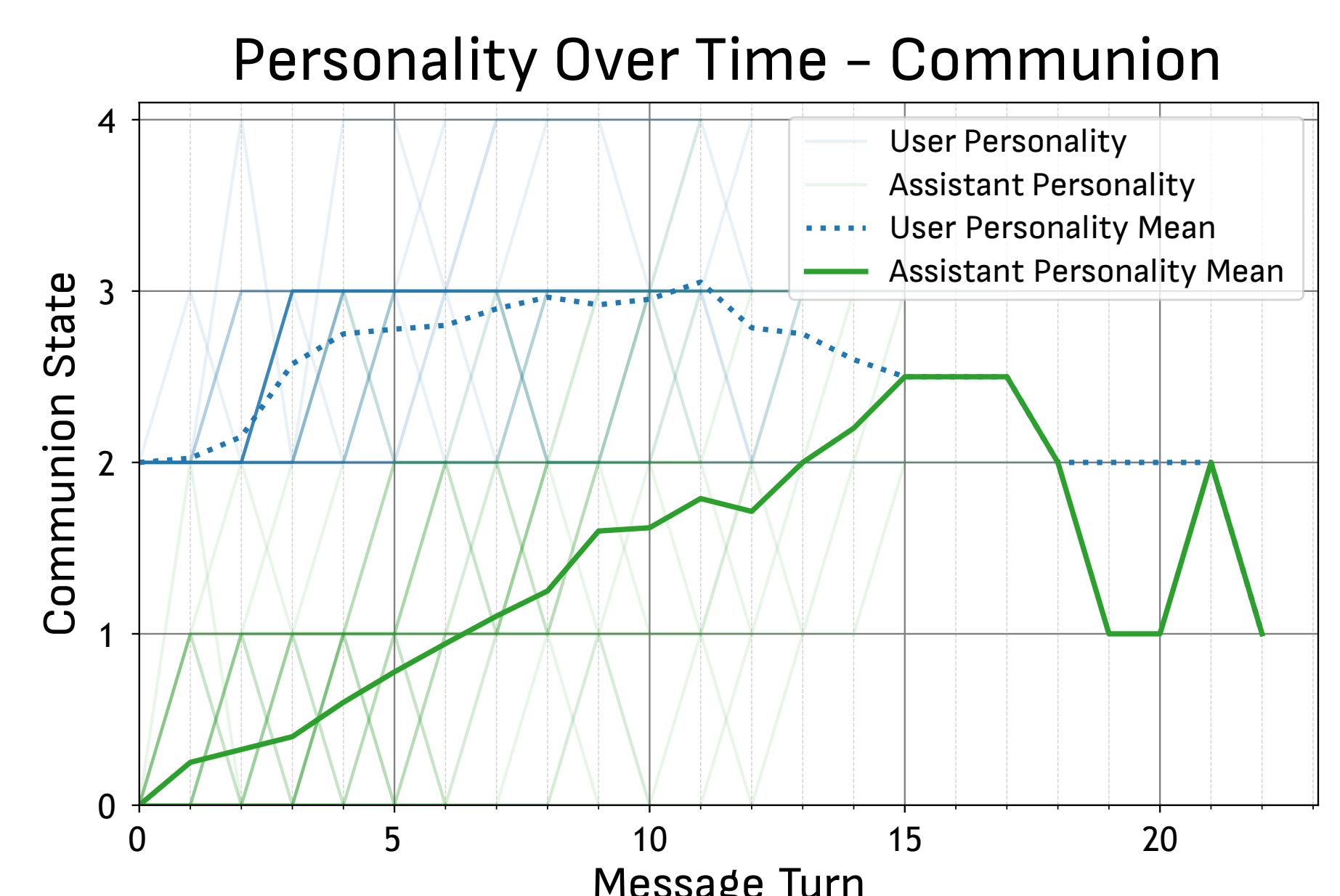
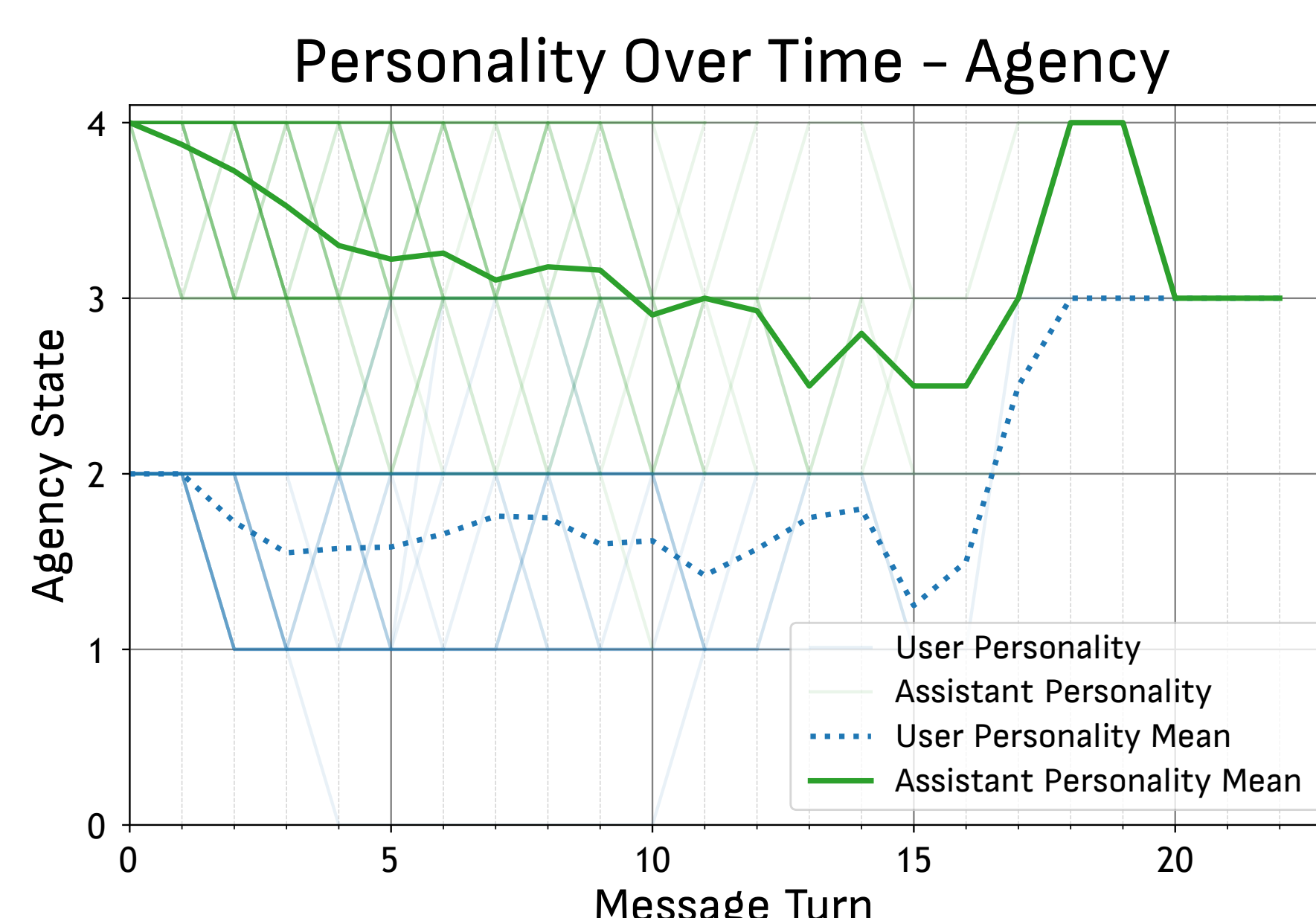
## Method



Agency  Communion 

## Results

- User study (N=40)
- Scenario: A dominant and uncooperative patient seeking a new appointment
- System adapts to user input
- Fine-tuned small model (Qwen 2.5 0.5B) achieves comparable performance to much larger models



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